

Homework 9

BSTA 550

Directions

Please turn in [this homework on Sakai](#). Please submit your homework in pdf format. You can type your work on your computer or submit a photo of your written work or any other method that can be turned into a pdf. The Adobe Scan phone app is an easy way to scan photos and compile into a PDF. Please let me know if you greatly prefer to submit a physical copy. We can work out another way for you to turn in homework.

Try to complete all of the problems listed below at some point this quarter! You may want to save some of them for studying later! Only turn in the ones listed in the “Turn In” column. Please submit problems in the order they are listed.

You must show all of your work to receive credit.

Chapter	Turn In	Extra Problems
36	TB # 12 ¹ , 14	TB # 4, 11, 13, 15, 16
37	TB # 24, 30	TB # 2, 4, 13, 20, 29
43	NTB # 1-2	TB # 1-4, 9 ² , 10 ³ , 11, 12 ⁴ NTB # 3-4

Non-textbook problems (NTB)

1. Let $R_X(t) = \ln(M_X(t))$. Show that $\text{Var}(X) = R_X''(0)$.

¹Assume the distances between the cars are independent.

²Include in your answer an explanation as to why we need the condition that $t < \lambda$.

³Do parts (a)-(c) below for #10 and #12: (a) Answer the question using the mgf $M_X(t)$ as instructed in the book. (b) Answer the question using $R_X(t)$ (as defined in class). (c) Which method did you prefer? Why?

⁴Do parts (a)-(c) below for #10 and #12: (a) Answer the question using the mgf $M_X(t)$ as instructed in the book. (b) Answer the question using $R_X(t)$ (as defined in class). (c) Which method did you prefer? Why?

2. The mgf for a Gamma distribution is $M_X(t) = \frac{1}{(1-t/\lambda)^r}$. Use the mgf of an Exponential distribution (from #43.9), to show that the sum of n i.i.d. $\text{Exponential}(\lambda)$ random variables has a $\text{Gamma}(r, \lambda)$ distribution.

Extra problems

3. Using mgf's, show that the sum of n i.i.d. Chi Square random variables with one degree of freedom ($\chi^2_{(1)}$) r.v.'s has a Chi Square with n degrees of freedom ($\chi^2_{(n)}$) distribution.

Hint: First, look up the pdf of a $\chi^2_{(n)}$. This is a special case of the Gamma distribution with what parameters? Based on that and the information from # 2 above, you can determine what the mgf of a $\chi^2_{(n)}$ is, which will help you determine whether the mgf of the sum of n i.i.d. $\chi^2_{(1)}$ r.v.'s has a $\chi^2_{(n)}$ distribution.

4. Use the mgf of a Poisson distribution to find the mgf of the following distributions. If the mgf is that of a common named distribution, then name the distribution and state its parameter(s).

1. The distribution of $\sum_{i=1}^n X_i$, if $X_i \sim \text{Poisson}(\lambda_i)$ and are independent.
2. The distribution of $\sum_{i=1}^3 X_i$, if $X_i \sim \text{Poisson}(\lambda)$ and are independent (i.i.d. in this case).
3. The distribution of $3X$, if $X \sim \text{Poisson}(\lambda)$.
4. Why are the answers to (b) and (c) different?

Some select answers

- Chapter 36
 - #4: 0.0044
 - # 12: (a) 0.9525 (b) 0.7939 (c) 0.7939
 - # 14: 0.5911
 - # 16:
 - * (a) $R = 8.225\sigma + 25\mu$
 - * (b) $R = 16.45\sigma + 100\mu$
 - * (c) $R = 164.5\sigma + 10,000\mu$
 - * (d) $R = 1.645\sqrt{n}\sigma + n\mu$

- Chapter 37
 - #2: 0.8869
 - # 4: 0.0023
 - # 20: 0.3936
 - # 24: 0.4562
 - # 30: (b) 0.0022 (c) $478.696 \approx 479$
- Chapter 43
 - NTB # 3:
 - * (a) $\text{Poisson}(\sum_{i=1}^n \lambda_i)$
 - * (b) $\text{Poisson}(3\lambda)$
 - * (c) $M_{3X}(t) = e\lambda(e^{3t} - 1)$ This is not an mgf of a common probability distribution.
 - * (d) In (b) we are adding independent r.v.'s X_i , while in (c) we are adding dependent r.v.'s ($3X = X + X + X$; X is dependent with itself).